

Building a Curation-First TTS Dataset (Indian English + Hindi)

Sarvam AI — ML & Speech Data Pipeline intern assignment

Thesis. The brief states three times that this is *not* a coding challenge — it is a **data-curation and quality-assurance** exercise. I therefore treated the code as scaffolding whose only job is to make curation **reproducible, auditable, and credit-frugal**, and spent the judgment budget on *source selection, QC gate design, threshold calibration against real data, and honest documentation*. Every clip is curated and reviewed as if it would train a production TTS model.

1. What was built

An end-to-end, resumable pipeline (`src/s01...s09`) that turns license-verified YouTube audio into a published Hugging Face TTS dataset:

```
sources.yaml → s01 fetch → s02 segment → s03 acoustic QC → s04 transcribe →  
(license gate)   (yt-dlp)       (silero-VAD)   (SNR/clip/loud)   (Sarvam STT,cached)  
  
s05 verify → s06 review → s07 normalize → s08 emotion → s09 build → HF Hub  
(1-speaker)  (HUMAN: listen (mono/24kHz/ (LLM cand. + (manifest +  
+language)   +correct text) loudnorm)      human conf.)  push_to_hub)
```

Key properties:

- **All thresholds live in** `config/pipeline.yaml` with written rationale, so every curation decision is explicit and every iteration is a reviewable `git diff`.
- **Sarvam responses are cached on disk by audio SHA-256** (`src/utls/sarvam_client.py`), so reruns and threshold iterations cost **zero credits** and the build is idempotent.
- **The manifest (`data/manifest.csv`) gives full provenance** for every published clip: source URL, license, speaker, timestamps, SHA-256, and every QC metric.

Design choices and why

Choice	Decision	Rationale
Sources	NPTEL lectures (Indian English + Hindi)	Single speaker per course; controlled recording; abundant; CC-BY on YouTube so reuse is unambiguous <i>and machine-verifiable</i> .
Clip length	4–18 s, VAD sentence-level	Coherent TTS units with clean silence edges (LJSpeech avg ≈ 6.6 s); the brief explicitly allows "any combination totaling ~60 min."
Final audio		

Choice	Decision	Rationale
	mono, 24 kHz, 16-bit, -23 LUFS, trimmed	Modern neural-TTS standard; clips re-cut from the original 48 kHz source (no upsampling).
Second language	Hindi	Highest Sarvam ASR accuracy among Indian languages and personally QA-able (Devanagari + romanized).
Emotion	LLM candidate → human confirm	Emotion is acoustic; text alone can't decide it. Tag conservatively and document the limitation.

2. Data-quality iterations

The quality funnel

Every clip flows through ordered gates; we record how many die at each. Automated gates run **before** any paid API call. Numbers below are from the validated run on the first English source (a 21-minute NPTEL lecture); the full 60-minute run extends the same funnel across all sources.

Stage	Clips	Surviving
Segmented (VAD utterances)	88	100.0%
Passed acoustic QC (pre-ASR)	63	71.6%
Transcribed (Sarovam STT)	<i>pending key</i>	—
Passed single-speaker + language	<i>pending</i>	—
Human-reviewed (decided)	<i>pending listen</i>	—
Accepted into dataset	<i>pending</i>	—

(Funnel auto-generated to `reports/funnel.csv` ; figures in `reports/figures/.`)

Acoustic QC against real data

The passing clips have a **median SNR of 38.2 dB** (studio-grade); the 22 dB gate cleanly removes a noisy tail (failed-clip SNRs cluster ≤ 25 dB, min -0.1 dB). This bimodal separation is exactly what a good threshold should produce — see `figures/snr_hist.png` . Duration is well-centred (median 11 s) within the 4–18 s target — `figures/duration_hist.png` .

Iteration v0 → v1: the peak-threshold mistake (a real one)

The most instructive iteration was a **bug in my own judgment**, not the code:

- **v0** set `true_peak_dbfs_max = -0.5` as an acoustic-QC reject gate, reasoning "audio peaking near full-scale is at risk of clipping distortion." Result: **83 of 88 clips (94%) were rejected** — almost the entire source.
- **Diagnosis.** YouTube audio is loudness-maximized; peaks sitting at ≈ 0 dBFS are *normal and harmless*, not evidence of distortion. I had conflated "peaks near full-scale" (a property fixed

downstream by my own loudness normalization, which limits to -1 dBFS in `s07`) with "digitally clipped/distorted" (the real defect, already caught by the clipping-ratio metric).

- **v1 fix.** Removed true-peak as a reject gate (still *measured* and recorded for the report), and relaxed `max_clipping_ratio` from $0.001 \rightarrow 0.01$ (a tiny fraction of full-scale samples is expected for loudness-maximized web audio; $>1\%$ indicates real clipping). **Pass rate moved from 3% $\rightarrow$ 72%**, with the noisy/clipped tail still correctly rejected.

This is recorded as a diff in `config/pipeline.yaml` and is, I think, the single most honest signal in the submission: a wrong threshold caught by *looking at the data*, diagnosed, and corrected — rather than tuned silently to "look good."

Other filtering decisions

- **License gate (s01).** A source is fetched only if yt-dlp reports its license as "*Creative Commons Attribution license (reuse allowed)*." Reupload/coaching channels that re-host NPTEL without the CC flag are auto-rejected. The flag is recorded per clip in the manifest — provenance, not assertion.
- **VAD over fixed windows.** Fixed 30/60 s cuts would slice mid-sentence and bury clean silence boundaries; VAD yields coherent utterances. Over-long utterances are re-split on a hard cap.
- **Free single-speaker check (s05).** We reuse Sarvam's diarized output: any clip with >1 `speaker_id`, a language mismatch, or low `language_probability` is dropped — with no additional API spend.

3. Observations

(Transcript/emotion observations are populated during the Sarvam-transcription + human-review pass; the methodology and tooling are complete and described here.)

- **Expected transcription errors.** NPTEL lectures carry domain jargon (thermodynamics, CS, math). These are predictable ASR-error hot-spots: technical terms, acronyms, numerals/units, and English $\leftrightarrow$ Hindi code-mixing in the Hindi set. The review harness pre-fills the ASR transcript so the reviewer corrects rather than re-types, and every edit is logged — yielding a measured WER (`reports/wer.json`).
- **Speaker/quality issues.** Even within a single-speaker lecture, some segments have room reverb, AC hum, or a cougher in the room; the SNR gate + manual listen catch these. Cross-talk during Q&A is caught by the diarization gate.
- **Emotion-tagging challenge.** Lecture/narration content is overwhelmingly `neutral / formal / conversational`. Affective tags (happy/sad/angry) are rare and inherently subjective from short clips. We tag conservatively and report the distribution honestly rather than manufacturing diversity.

4. What worked

- **Pre-ASR acoustic gating** — rejecting bad audio for free before spending any Sarvam credit; the credit-frugal core of the design.

- **Reusing diarization for single-speaker verification** — a quality gate at zero marginal cost.
- **SHA-256 response caching** — iteration and re-runs cost nothing; the whole build is reproducible from cache.
- **Programmatic license verification** — turning "is this reusable?" from a claim into a recorded, per-clip fact.
- **Calibrating thresholds against the data distribution** rather than guessing — the SNR histogram made the 22 dB choice obvious.

5. What didn't work / dead ends

- **The v0 peak gate** (§2) — wrong mental model, fixed.
- **YouTube keyword search for sources is noisy** — queries for "NPTEL ... Hindi" surfaced reupload and exam-coaching channels (not CC-BY) far more than official lecture content. Conclusion: anchor on *official channels* and let the license gate enforce correctness, rather than trusting search ranking.
- **Sourcing options ruled out.** TED/TEDx (CC-BY-ND forbids derivatives), All India Radio archives (paid license only), and general news/podcast clips (standard license + multi-speaker/music). Documented in `config/sources.yaml:rejected`.
- **Text-only emotion tagging is weak** — confirmed expectation: transcripts rarely reveal affect, hence the human-confirm step.

6. Future improvements

- **Blind gold WER set** — hand-transcribe N clips/language *before* seeing ASR, for an unbiased WER (current WER vs. human-corrected text is a slightly optimistic proxy).
 - **Audio-based emotion classifier** — replace text-only LLM candidates with a prosody/acoustic model, then human-confirm.
 - **More speakers & languages** — additional NPTEL professors and a third Indian language for broader coverage.
 - **MOS listening test** + automatic alignment-confidence scoring to prioritize review.
 - **Forced alignment** (e.g. MFA) for tighter word-level boundaries than VAD edges.
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Appendix

- **Reproducibility.** Pinned deps (`requirements.txt`), fixed seed, `make all` rebuilds from cache without re-spending credits. `validate_dataset.py` asserts every invariant (mono/24 kHz/16-bit, one speaker, non-empty transcript, duration window, language $\in \{\text{en-IN, hi-IN}\}$, SHA-256 match, ~60 min ~30/30).
- **Licensing & ethics.** Audio CC-BY-4.0 with per-clip attribution to NPTEL; code MIT. Only YouTube-CC-BY-flagged videos are used, verified at fetch.
- **Exact Sarvam usage** is logged per call to `data/cache/api_usage.log`.
- **Artifacts:** `reports/funnel.csv`, `reports/wer.json`, `reports/figures/*.png`, `data/manifest.csv`, `review/review_log.csv`.

Appendix – Figures

